

Local duality and structure of the $H^i(M)$

SGA 2, Exposé V

English draft

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About this draft

This is an unofficial English translation of Exposé V, “Local duality and structure of the $H^i(M)$ ”, from SGA 2. The source volume is A. Grothendieck (notes by a group of auditors), with an exposé by M. Raynaud, *Cohomologie locale des faisceaux cohérents et théorèmes de Lefschetz locaux et globaux* (SGA 2), Séminaire de géométrie algébrique du Bois Marie, 1962. It follows the slightly corrected SMF recomposition, arXiv:math/0511279.

This draft contains the assigned body of the source, including proofs, original footnotes, and editor notes (N.D.E.). Statement numbering and mathematical notation follow the source. References to material outside this draft retain their original numbers. Apparent mathematical misprints in the source have been retained; they are recorded in the accompanying README, separately from the translated text. Scholarly proofreading remains outstanding.

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EXPOSÉ V

Local duality and structure of the $H^i(M)$

1. Complexes of homomorphisms

1.1. Let $F^\bullet$ and $G^\bullet$ be two graded modules; one then denotes by:

$$(1) \quad \text{Hom}^\bullet(F^\bullet, G^\bullet)$$

the graded module of homomorphisms of graded modules from $F^\bullet$ to $G^\bullet$. Thus one has:

$$(2) \quad \text{Hom}^s(F^\bullet, G^\bullet) = \prod_k \text{Hom}(F^k, G^{k+s}).$$

Let $F^\bullet$ (resp. $G^\bullet$) be a complex, and let d_1 (resp. d_2) be its differential; then for $h \in \text{Hom}^s(F^\bullet, G^\bullet)$ one sets¹

$$(3) \quad d(h) = h \circ d_1 + (-1)^{s+1} d_2 \circ h.$$

One verifies trivially that $d \circ d = 0$, hence that $\text{Hom}^\bullet(F^\bullet, G^\bullet)$ equipped with d is a complex. The cohomology group of this complex is denoted

$$(4) \quad \mathbb{H}^\bullet(F^\bullet, G^\bullet).$$

If $G^\bullet$ is injective in each degree, then

$$F^\bullet \longmapsto \mathbb{H}^\bullet(F^\bullet, G^\bullet)$$

is an exact ∂ -functor. Likewise, for arbitrary $F^\bullet$,

$$G^\bullet \longmapsto \mathbb{H}^\bullet(F^\bullet, G^\bullet)$$

is an exact δ -functor on the category of complexes $G^\bullet$ injective in each degree.

REMARK 1.2. The cycles of $\text{Hom}^\bullet(F^\bullet, G^\bullet)$ are the homomorphisms from $F^\bullet$ to $G^\bullet$ which commute or anticommute, according to the degree, with the differentials. The boundaries of $\text{Hom}^\bullet(F^\bullet, G^\bullet)$ are the homomorphisms from $F^\bullet$ to $G^\bullet$ homotopic to zero.

Let A be a ring and let M (resp. N) be an A -module, $R(M)$ (resp. $R(N)$) an injective resolution of M (resp. N); then there exists a canonical isomorphism²

$$(1.3) \quad \mathbb{H}^s(R(M), R(N)) \simeq \text{Ext}^s(M, N).$$

Indeed, let $i: M \rightarrow R(M)$ be the canonical augmentation, and let $h \in \text{Hom}^s(R(M), R(N))$; one then denotes by t_s the map

$$h \longmapsto h_0 \circ i$$

from $\text{Hom}^s(R(M), R(N))$ to $\text{Hom}(M, R(N)^s)$. The family $(t_s)_{s \geq 0}$ defines a homomorphism of (ordinary) complexes³

$$t: \text{Hom}^\bullet(R(M), R(N)) \longrightarrow \text{Hom}^\bullet(M, R(N)),$$

i.e. one has $(dh)_0 \circ i = d_2 \circ h_0 \circ i$.

¹**N.D.E.** the original sign convention was different; but it is not compatible with the convention of Exposé ??, which seems more reasonable, since in that case the cohomology in degree 0 is then the set of homotopy classes of morphisms from $F^\bullet$ to $G^\bullet$. The calculations have been modified accordingly in what follows.

²**N.D.E.** one has retained the original's strange numbering.

³**N.D.E.** one still denotes by M the complex $M[0]$ consisting of M placed in degree 0.

One verifies easily that, passing to cohomology, t gives an isomorphism. In particular it follows that

$$\underset{\sim}{\mathrm{H}}^{\bullet}(R(M), R(N))$$

does “not depend” on the chosen injective resolution $R(M)$ (resp. $R(N)$) of M (resp. N).

To every exact sequence of A -modules

$$(5) \quad 0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

one associates an exact sequence of injective resolutions

$$(6) \quad 0 \longrightarrow R(M') \longrightarrow R(M) \longrightarrow R(M'') \longrightarrow 0.$$

One verifies that the isomorphism (1.3) commutes with the homomorphisms:

$$(8) \quad \underset{\sim}{\mathrm{H}}^s(R(M'), R(N)) \longrightarrow \underset{\sim}{\mathrm{H}}^{s+1}(R(M''), R(N)),$$

$$(9) \quad \mathrm{Ext}^s(R(M'), R(N)) \longrightarrow \mathrm{Ext}^{s+1}(R(M''), R(N)),$$

deduced from (6) and (5).

Let P be a third A -module, $R(P)$ an injective resolution of P ; then composition of graded morphisms yields a pairing

$$(10) \quad \mathrm{Hom}^i(R(N), R(M)) \times \mathrm{Hom}^j(R(M), R(P)) \longrightarrow \mathrm{Hom}^{i+j}(R(N), R(P)),$$

which defines a pairing:

$$(11) \quad \underset{\sim}{\mathrm{H}}^i(R(N), R(M)) \times \underset{\sim}{\mathrm{H}}^j(R(M), R(P)) \longrightarrow \underset{\sim}{\mathrm{H}}^{i+j}(R(N), R(P)),$$

hence a homomorphism of functors in M :

$$(1.4) \quad \underset{\sim}{\mathrm{H}}^i(R(N), R(M)) \longrightarrow \mathrm{Hom}(\underset{\sim}{\mathrm{H}}^j(R(M), R(P)), \underset{\sim}{\mathrm{H}}^{i+j}(R(N), R(P))).$$

One will see that (1.4) is a homomorphism of δ -functors in M . The exact sequences (5) and (6) yield a commutative diagram:

$$\begin{array}{ccc} \mathrm{Hom}^i(R(N), R(M')) & \longrightarrow & \mathrm{Hom}(\mathrm{Hom}^j(R(M'), R(P)), \mathrm{Hom}^{i+j}(R(N), R(P))) \\ \downarrow & & \downarrow \mathrm{Hom}(q, \mathrm{id}) \\ \mathrm{Hom}^i(R(N), R(M)) & \longrightarrow & \mathrm{Hom}(\mathrm{Hom}^j(R(M), R(P)), \mathrm{Hom}^{i+j}(R(N), R(P))) \\ \downarrow p & & \downarrow \\ \mathrm{Hom}^i(R(N), R(M'')) & \longrightarrow & \mathrm{Hom}(\mathrm{Hom}^j(R(M''), R(P)), \mathrm{Hom}^{i+j}(R(N), R(P))). \end{array}$$

Let $h \in \mathrm{Hom}^i(R(N), R(M''))$ (resp. $g \in \mathrm{Hom}^j(R(M'), R(P))$) be a cycle, and let $h' \in \mathrm{Hom}^i(R(N), R(M))$ (resp. $g' \in \mathrm{Hom}^j(R(M), R(P))$) be such that $p(h') = h$ (resp. $q(g') = g$); then to say that (1.4) is a homomorphism of δ -functors in M is to say that

$$(12) \quad g \circ dh' - dg' \circ h$$

is a coboundary in $\mathrm{Hom}^{\bullet}(R(N), R(P))$.

Now one has:

$$\begin{aligned} dh' &= h' \circ d_1 + (-1)^{i+1} d_2 \circ h' \\ dg' &= g' \circ d_2 + (-1)^{j+1} d_3 \circ g' \end{aligned}$$

with the obvious notation. Thus (12) may be written:

$$g \circ h' \circ d_1 + (-1)^{i+1} g \circ d_2 \circ h' - g' \circ d_2 \circ h - (-1)^{j+1} d_3 \circ g' \circ h.$$

On the other hand, since h and g are cycles one has:

$$\begin{aligned} g \circ d_2 &= (-1)^j d_3 \circ g, \\ d_2 \circ h &= (-1)^i h \circ d_1, \end{aligned}$$

hence, finally, (12) may be written:

$$d(g \circ h' + (-1)^{i+1} g' \circ h),$$

which concludes the proof.

(1.3) and (1.4) thus yield a homomorphism of δ -functors in M :

$$(1.5) \quad \text{Ext}^i(N, M) \longrightarrow \text{Hom}(\text{Ext}^j(M, P), \text{Ext}^{i+j}(N, P)).$$

2. The local duality theorem for a regular local ring

Let A be a regular local ring of dimension r , $\mathfrak{m}$ the maximal ideal of A and M an A -module of finite type. One sets $H^i(M) = H_{\mathfrak{m}}^i(M)$ (hence $H^i(M) = \varinjlim \text{Ext}^i(A/\mathfrak{m}^k, M)$). One has seen (?? ??) that $I = H^r(A)$ is a dualizing module for A ; let D denote the associated dualizing functor. In (1.5) set $N = A/\mathfrak{m}^k$, $P = A$; one then obtains a homomorphism of δ -functors in M

$$(13) \quad \varphi_k: \text{Ext}^i(A/\mathfrak{m}^k, M) \longrightarrow \text{Hom}(\text{Ext}^{r-i}(M, A), \text{Ext}^r(A/\mathfrak{m}^k, A)).$$

Passing to the inductive limit with respect to k , one finds a homomorphism of δ -functors

$$(14) \quad \varphi: H^i(M) \longrightarrow D(\text{Ext}^{r-i}(M, A)).$$

THEOREM 2.1 (Local duality theorem). *The preceding functorial homomorphism φ is an isomorphism.*

PROOF. If $i > r$, the second member of (14) is trivially zero, and the first member is zero since $H^i(M) = \varinjlim_k \text{Ext}^i(A/\mathfrak{m}^k, M)$, and this is true for each $\text{Ext}^i(A/\mathfrak{m}^k, M)$ (syzygy theorem).

If $i = r$, by the preceding, the two functors in M , $H^r(M)$ and $D(\text{Hom}(M, A))$ are right exact; since A is noetherian, and M of finite type, it suffices to check the isomorphism for $M = A$, which is immediate.

To show that φ is a functorial isomorphism, it now suffices, proceeding by descending induction with respect to i , to note that every module of finite type admits a finite presentation, and that for $i < r$ both members of (14) are zero if M is free of finite type. This is obvious for the second member, and since H^i commutes with finite sums it suffices, as for the first, to show that $H^i(A) = 0$ for $i < r$. Now this follows, since $\text{prof}(A) = r$, from (?? ??). $\square$

3. Application to the structure of the $H^i(M)$

THEOREM 3.1. *Let A be a noetherian local ring, D a dualizing functor for A , M an A -module of finite type $\neq 0$, and of dimension n , then one has:*

- (i) $H^i(M) = 0$ if $i < 0$ or if $i > n$.
- (ii) $D(\widehat{H^i(M)})$ is a module of finite type over $\widehat{A}$, of dimension $\leq i$.
- (iii) $H^n(M) \neq 0$, and if A is complete, $D(H^n(M))$ is of dimension n and $\text{Ass}(D(H^n(M))) = \{\mathfrak{p} \in \text{Ass}(M) \mid \dim A/\mathfrak{p} = n\}$.

PROOF. Let I be the dualizing module associated to D . One knows that $\widehat{I}$ is a dualizing module for $\widehat{A}$. On the other hand, one has:

$$\begin{aligned} \widehat{H^i(M)} &= H^i(\widehat{M}), \\ D(\widehat{H^i(M)}) &= \text{Hom}(H^i(\widehat{M}), \widehat{I}) \text{ and} \\ \dim \widehat{M} &= \dim M, \end{aligned}$$

hence one may assume A complete. Now, by a theorem of Cohen, every complete local ring is a quotient of a regular local ring. To reduce to this case, one needs the following lemma:

LEMMA 3.2. *Let X (resp. Y) be a ringed space, X' (resp. Y') a closed subspace of X (resp. Y), and $f: X \rightarrow Y$ a morphism of ringed spaces such that $f^{-1}(Y') = X'$. Let F be an $\mathcal{O}_X$ -Module and denote by A (resp. B) the ring $\Gamma(\mathcal{O}_X)$ (resp. $\Gamma(\mathcal{O}_{Y'})$) and by $\bar{f}: B \rightarrow A$ the ring homomorphism corresponding to f . There exists a spectral sequence of B -modules, with initial term*

$$(15) \quad E_2^{p,q} = H_{Y'}^p(Y, R^q f_*(F)),$$

abutting to the B -module $H_{X'}^\bullet(X, F)_{[\bar{f}]}$.

PROOF. Let $\mathcal{O}_{Y,Y'}$ be the sheaf $\mathcal{O}_Y|_{Y'}$ extended by 0 outside of Y' (see Exp. ??). One has an isomorphism of B -modules:

$$(16) \quad \text{Hom}(\mathcal{O}_{Y,Y'}, f_*(F)) \simeq \text{Hom}(f^*(\mathcal{O}_{Y,Y'}), F)_{[\bar{f}]}.$$

Now, one has:

$$(17) \quad f^*(\mathcal{O}_{Y,Y'}) = \mathcal{O}_{X,X'},$$

and moreover if G is an injective $\mathcal{O}_X$ -Module, then $f_*(G)$ is an injective $\mathcal{O}_Y$ -Module, at least if f is flat, a case to which one can easily reduce by replacing $\mathcal{O}_X$ etc. by the constant sheaves of rings $\mathbf{Z}$. Hence the spectral sequence of the composed functor

$$F \longrightarrow \text{Hom}(\mathcal{O}_{Y,Y'}, f_*(F)),$$

with initial term

$$E_2^{p,q} = \text{Ext}^p(Y; \mathcal{O}_{Y,Y'}, R^q f_*(F)),$$

abuts, taking (16) and (17) into account, to:

$$\text{Ext}^\bullet(X; \mathcal{O}_{X,X'}, F)_{[\bar{f}]}.$$

The lemma then follows from (?? ?? bis). $\square$

Now let $f: B \rightarrow A$ be a surjective homomorphism of local rings. Let

$$f: \text{Spec}(A) \longrightarrow \text{Spec}(B)$$

be the corresponding morphism of affine schemes. Set $X = \text{Spec}(A)$ (resp. $X' = \{\mathfrak{m}_A\}$), $Y = \text{Spec}(B)$ (resp. $Y' = \{\mathfrak{m}_B\}$), and let M be an A -module and $\widetilde{M}$ the corresponding $\mathcal{O}_X$ -Module. Since $R^q f_*(\widetilde{M}) = 0$ for $q > 0$, the spectral sequence (15) degenerates, and one obtains by (3.2) an isomorphism of B -modules:

$$(18) \quad H_{\{\mathfrak{m}_B\}}^n(Y, f_*(\widetilde{M})) \simeq H_{\{\mathfrak{m}_A\}}^n(X, \widetilde{M})_{[f]},$$

hence an isomorphism of B -modules:

$$(19) \quad H_{\mathfrak{m}_B}^n(M_{[f]}) \simeq H_{\mathfrak{m}_A}^n(M)_{[f]}.$$

On the other hand if D_A (resp. D_B) is the dualizing functor for A (resp. B), one has:

$$(20) \quad D_A(M)_{[f]} \simeq D_B(M_{[f]}).$$

Finally, since one has a ring isomorphism

$$(21) \quad B / \text{Ann } M_{[f]} \simeq A / \text{Ann } M,$$

one sees that the change of base rings under consideration changes nothing. Suppose therefore that A is regular of dimension r .

By (2.1) one has:

$$(22) \quad D(H^i(M)) = \text{Ext}^{r-i}(M, A).$$

One will prove the equivalence of the following properties:

- (a) $\dim \text{Ext}^j(M, A) \leq r - j$;
- (b) for every $\mathfrak{p} \in X = \text{Spec}(A)$ such that $\dim A_{\mathfrak{p}} < j$, one has $\text{Ext}^j(M, A)_{\mathfrak{p}} = 0$;
- (c) $\text{codim}(\text{Supp}(\text{Ext}^j(M, A)), X) \geq j$.

To prove (a) $\Rightarrow$ (b), let $\mathfrak{p} \in X$, $\dim A_{\mathfrak{p}} < j$, then $\dim A/\mathfrak{p} > r - j$, hence by (a) $\text{Ann}(\text{Ext}^j(M, A)) \not\subset \mathfrak{p}$, which implies $\text{Ext}^j(M, A)_{\mathfrak{p}} = 0$. Let $\mathfrak{p} \in \text{Supp}(\text{Ext}^j(M, A))$ then $\text{Ext}^j(M, A)_{\mathfrak{p}} \neq 0$ hence by (b) $\dim A_{\mathfrak{p}} \geq j$. Hence $\text{codim}(\text{Supp}(\text{Ext}^j(M, A)), X) = \inf\{\dim A_{\mathfrak{p}} \mid \mathfrak{p} \in \text{Supp}(\text{Ext}^j(M, A))\} \geq j$, that is (b) $\Rightarrow$ (c). Finally (c) implies (a) trivially.

Let us now prove the theorem.

- (i) Let $x = (x_1, \dots, x_r)$ be a system of parameters for A such that $x_i \in \text{Ann } M$ for $i = 1, \dots, r - n$. Let $K^\bullet((x^k), M)$ be the Koszul complex. One sees easily that the map $K^i((x^k), M) \rightarrow K^i((x^{k'}), M)$ for $k < k'$ is zero, if $i > n$. It follows that $H^i(M) = \varinjlim H^i((x^k), M) = 0$ if $i > n$. On the other hand, it is trivial that $H^i(M) = 0$ if $i < 0$, hence (i) is proved.
 - (ii) Since A is regular, $\dim A_{\mathfrak{p}} < j$ implies that the global homological dimension of $A_{\mathfrak{p}}$ is strictly smaller than j and hence $\text{Ext}^j(M, A)_{\mathfrak{p}} = \text{Ext}_{A_{\mathfrak{p}}}^j(M_{\mathfrak{p}}, A_{\mathfrak{p}}) = 0$, hence one has proved (b) and consequently (a). (ii) then follows from (22) and from (a).
 - (iii) There exists a $\mathfrak{p} \in \text{Supp}(M)$ such that $\dim A_{\mathfrak{p}} = r - n$ and such that $\text{Supp}(M_{\mathfrak{p}}) = \{\mathfrak{m}_{A_{\mathfrak{p}}}\}$. Since $A_{\mathfrak{p}}$ is regular if A is, one finds $\text{prof } A_{\mathfrak{p}} = r - n$ hence
- $$(23) \quad \text{Ext}_A^{r-n}(M, A)_{\mathfrak{p}} = \text{Ext}_{A_{\mathfrak{p}}}^{r-n}(M_{\mathfrak{p}}, A_{\mathfrak{p}}) \neq 0.$$

This implies, taking (22) into account, that on the one hand:

$$H^n(M) \neq 0,$$

on the other hand

$$\dim D(H^n(M)) \geq n,$$

hence by (ii)

$$\dim D(H^n(M)) = n.$$

Now let $Y = \text{Supp}(M)$. By (i) one knows that $D(H^n(M')) = \text{Ext}^{r-n}(M', A)$ is a functor in M' , left exact, on the category $(\mathbf{C}_Y)^\circ$. Hence there exists an A -module H and an isomorphism of functors in M' :

$$\text{Ext}^{r-n}(M', A) = \text{Hom}(M', H).$$

Let Y_i , $i = 1, \dots, k$ be the irreducible components of Y of maximum dimension. One will see that the assertion $\text{Ext}^{r-n}(M', A) \neq 0$ is equivalent to the assertion: there exists an i such that $\text{Supp } M' \supset Y_i$. Indeed if $\text{Supp } M' \supset Y_i$ then $\dim(M') = n$ hence $\text{Ext}^{r-n}(M', A) \neq 0$.

If $\text{Supp } M' \not\supset Y_i$ for every $i = 1, \dots, k$, then $\dim M' < n$

$$D(H^n(M')) = \text{Ext}^{r-n}(M', A) = 0.$$

Since $\text{Ass}(\text{Ext}^{r-n}(M, A)) = \text{Supp } M \cap \text{Ass}(H)$ one sees that the last assertion of (iii) follows from the following lemma:

LEMMA 3.3. *Let $X = \text{Spec}(A)$, and let Y be a closed subset of X , $T: (\mathbf{C}_Y)^\circ \rightarrow \underline{\text{Ab}}$ a left exact functor, and Y_i , $i = 1, \dots, k$ a family of irreducible components of Y such that the assertion: $T(M) = 0$ is equivalent to the assertion: $\forall i \text{ Supp } M \not\supset Y_i$. Then T is representable by a module H such that $\text{Ass}(H) = \bigcup_{i=1}^k \{y_i\}$, where y_i is the generic point of Y_i , $i = 1, \dots, k$.*

PROOF. Let $y \in Y$, one constructs an A -module $M(y)$ such that $\text{Supp}(M(y)) = \overline{\{y\}}$. Suppose that $y \neq y_i$ for every $i = 1, \dots, k$, then $Y_i \not\subset \text{Supp}(M(y))$ for every $i = 1, \dots, k$, hence $T(M(y)) = 0$. It follows:

$$\text{Ass}(T(M(y))) = \text{Supp}(M(y)) \cap \text{Ass}(H) = \emptyset,$$

hence $y \notin \text{Ass}(H)$. If $y = y_i$, then $Y_i \subset \text{Supp}(M(y))$, hence $T(M(y)) \neq 0$, hence:

$$\text{Ass}(T(M(y))) = \text{Supp}(M(y)) \cap \text{Ass}(H) \neq \emptyset.$$

By the first part of the proof, this implies $y \in \text{Ass}(H)$, whence the lemma, Q.E.D. $\square$

□

EXAMPLE 3.4. Let A be a noetherian ring, let $X = \text{Spec}(A)$, and Y a closed subset of X , such that $X - Y$ is affine, then for every irreducible component Y_α of Y one has $\text{codim}(Y_\alpha, X) \leq 1$.

Indeed, consider X as a prescheme over X . Let $y_\alpha \in Y_\alpha$ be a generic point, and consider the morphism $\text{Spec}(\mathcal{O}_{X, y_\alpha}) \rightarrow X$. The affine scheme obtained by extension of the base scheme of X to $\text{Spec}(\mathcal{O}_{X, y_\alpha})$ is canonically isomorphic to $\text{Spec}(\mathcal{O}_{X, y_\alpha})$.

By (EGA I 3.2.7) one sees that if y_0 is the unique closed point of $Y_0 = \text{Spec}(\mathcal{O}_{X, y_\alpha})$ then $Y_0 - y_0$ is affine. By (EGA III 1.3.1) one finds:

$$H^i(Y_0 - y_0, \mathcal{O}_{Y_0}) = 0 \quad \text{if } i > 0,$$

hence by (?? ??)

$$H^{i-1}(\mathcal{O}_{X, y_\alpha}) = H^i_{\{y_0\}}(Y_0, \mathcal{O}_{Y_0}) = 0 \quad \text{if } i \geq 2.$$

Taking 3.1 (iii) into account, it follows:

$$\dim \mathcal{O}_{X, y_\alpha} \leq 1,$$

$$\text{hence } \text{codim}(Y_\alpha, X) = \inf_{y \in Y_\alpha} \dim \mathcal{O}_{X, y} \leq 1,$$

Q.E.D.

Let A be a noetherian local ring, $\mathfrak{m}$ the maximal ideal and M an A -module of finite type. Suppose that A is a quotient of a regular local ring. Set $X = \text{Spec}(A)$, and for every $x \in X$, $\mathfrak{m}_x = \mathfrak{m}A_x$.

PROPOSITION 3.5. *The following two conditions are equivalent:*

- a) $H^i(M)$ is of finite length,
- b) $\forall x \in X - \{\mathfrak{m}\}, H_{\mathfrak{m}_x}^{i - \dim \{x\}}(M_x) = 0$.

PROOF. Taking (3.2) into account we may assume A regular. By (2.1) we have:

$$H^i(M) = D(\text{Ext}^{r-i}(M, A)),$$

where $r = \dim A$. By (?? ??), a) is equivalent⁴ to:

$$(24) \quad \text{Ext}^{r-i}(M, A) \text{ is of finite length.}$$

Now (24) is equivalent to:

$$(25) \quad \forall x \in X - \{\mathfrak{m}\}, \text{ one has } \text{Ext}^{r-i}(M, A)_x = 0.$$

On the other hand A_x is regular of dimension $r - \dim \overline{\{x\}}$, hence by (2.1)

$$(26) \quad H_{\mathfrak{m}_x}^{i - \dim \overline{\{x\}}}(M_x) = D(\text{Ext}_{A_x}^{(r - \dim \overline{\{x\}}) - (i - \dim \overline{\{x\}})}(M_x, A_x)) = D(\text{Ext}_{A_x}^{r-i}(M_x, A_x)).$$

Since M is of finite type one has:

$$\text{Ext}_A^{r-i}(M, A)_x = \text{Ext}_{A_x}^{r-i}(M_x, A_x)$$

whence the proposition. □

⁴N.D.E. indeed, it is a matter of showing that, E being an A -module of finite type, if $D(E)$ is of finite length then E is of finite length. Let K (resp. Q) be the kernel (resp. cokernel) of the canonical morphism $\epsilon : E \rightarrow DD(E)$. The composite of $D(\epsilon)$ and of the canonical morphism $\gamma : D(E) \xrightarrow{\gamma} DDD(E)$ is the identity of $D(E)$. As $D(E)$ is of finite length, γ is an isomorphism and hence $D(\epsilon)$ likewise. As D is exact, one deduces that $D(K)$ and $D(Q)$ vanish. It suffices to prove that if M is an A -module of vanishing dual, then M is zero, for one will then have $E = DD(E)$ of finite length like $D(E)$. Indeed, let M_0 be a submodule of finite type of M . As D is exact, $D(M_0)$ is a quotient of $D(M)$, which is therefore zero. Still by exactness of D , one has $D(M_0/\mathfrak{m}_A M_0) = 0$, and hence, by biduality, the module of finite length $M_0/\mathfrak{m}_A M_0$ is zero. Nakayama's lemma then ensures the vanishing of M_0 and finally one obtains that of M .

COROLLARY 3.6. *In order that $H^i(M)$ be of finite length for $i \leq n$, it is necessary and sufficient that*

$$\text{prof}(M_x) > n - \dim \overline{\{x\}}$$

for every $x \in X - \{\mathfrak{m}\}$.

PROOF. Follows from (3.5) and from (?? ??). □